

MATLAB ASSIGNMENT 5
BY GROUP 3 APE

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Newton Raphson

```
clc;
clear;

f = @(x) x.^3 - x - 2;
df = @(x) 3*x.^2 - 1;

x = 1.5;
tol = 1e-5;
k = 0;
maxIter = 20;

while k < maxIter
    xnew = x - f(x)/df(x);
    k = k + 1;

    fprintf('Iteration %d: x = %.6f\n', k, xnew);

    if abs(xnew - x) < tol
        x = xnew;
        break;
    end

    x = xnew;
end

fprintf('Root = %.6f\n', x);
```

```
clc;
clear;

f = @(x) x.^3 - x - 2;
df = @(x) 3*x.^2 - 1;

x = 1.5;
tol = 1e-5;
N = 20;

for k = 1:N
    xnew = x - f(x)/df(x);

    fprintf('Iteration %d: x = %.6f\n', k, xnew);

    if abs(xnew - x) < tol
        x = xnew;
        break;
    end

    x = xnew;
end

fprintf('Root = %.6f\n', x);
```

PSEUDOCODE

START

Define $f(x) = x^3 - x - 2$

Define $df(x) = 3x^2 - 1$

Set $x_0 = 1.5$

Set tolerance = 0.000001

Set maxiter = 20

Set $i = 0$

WHILE $i < \text{maxiter}$

$fx = f(x_0)$

$dfx = df(x_0)$

 IF $dfx == 0$ THEN

 Display "Derivative is zero"

 STOP

 END IF

$x_1 = x_0 - fx/dfx$

 error = $ABS(x_1 - x_0)$

 Display $i+1, x_0, x_1, \text{error}$

 IF error < tolerance THEN

 Display "Root =", x_1

 STOP

 END IF

$x_0 = x_1$

$i = i + 1$

END WHILE

Display "Maximum iterations reached"

END

Secant method

```
clc;
clear;

f = @(x) x.^3 - x - 2;

x0 = 1;
x1 = 2;
tol = 1e-5;
N = 20;

for k = 1:N
    x2 = x1 - f(x1)*(x1-x0)/(f(x1)-f(x0));

    fprintf('Iteration %d: x = %.6f\n', k, x2);

    if abs(x2-x1) < tol
        x1 = x2;
        break;
    end

    x0 = x1;
    x1 = x2;
end

fprintf('Root = %.6f\n', x1);
```

```
clc;
clear;

f = @(x) x.^3 - x - 2;

x0 = 1;
x1 = 2;
tol = 1e-5;
k = 0;
maxIter = 20;

while k < maxIter
    x2 = x1 - f(x1)*(x1-x0)/(f(x1)-f(x0));
    k = k + 1;

    fprintf('Iteration %d: x = %.6f\n', k, x2);

    if abs(x2-x1) < tol
        x1 = x2;
        break;
    end

    x0 = x1;
    x1 = x2;
end
```

PSEDOCODE

```
START
DEFINE f(x, y) = x + y
SET x0 = 0
SET y0 = 1
SET h = 0.1
SET x_end = 0.2

SET n = (x_end - x0) / h    // number of steps = 2
SET x = x0
SET y = y0
FOR i = 1 TO n
    K1 = h * f(x, y)
    K2 = h * f(x + h/2, y + k1/2)
    K3 = h * f(x + h/2, y + k2/2)
    K4 = h * f(x + h, y + k3)
    Y = y + (k1 + 2*k2 + 2*k3 + k4) / 6
    X = x + h
    PRINT i, x, k1, k2, k3, k4, y
END FOR
PRINT "y(", x_end, ") ≈ ", y
END
```

```

clc;
clear;

f = @(x,y) x + y;

x0 = 0;
y0 = 1;
h = 0.1;
xf = 0.2;

x = x0;
y = y0;

N = round((xf-x0)/h);

for n = 1:N
    k1 = h*f(x,y);
    k2 = h*f(x+h/2, y+k1/2);
    k3 = h*f(x+h/2, y+k2/2);
    k4 = h*f(x+h, y+k3);

    y = y + (k1+2*k2+2*k3+k4)/6;
    x = x + h;

    fprintf('Step %d: x = %.2f, y = %.6f\n', n, x, y);
end

fprintf('y(%.2f) = %.6f\n', xf, y);

```

```

clc;
clear;

f = @(x,y) x + y;

x = 0;
y = 1;
h = 0.1;
xf = 0.2;

n = 0;

while x < xf - 1e-12
    k1 = h*f(x,y);
    k2 = h*f(x+h/2, y+k1/2);
    k3 = h*f(x+h/2, y+k2/2);
    k4 = h*f(x+h, y+k3);

    y = y + (k1+2*k2+2*k3+k4)/6;
    x = x + h;
    n = n + 1;

    fprintf('Step %d: x = %.2f, y = %.6f\n', n, x, y);
end

fprintf('y(%.2f) = %.6f\n', xf, y);

```

PESUDOCODE

START

Define $f(x) = x^2 - x - 2$

Set $x_0 = 1$

Set $x_1 = 2$

Set tolerance = 0.00005

REPEAT

Calculate $f_0 = f(x_0)$

Calculate $f_1 = f(x_1)$

Calculate:

$x_2 = x_1 - [f_1 \times (x_1 - x_0)] / [f_1 - f_0]$

Calculate $f_2 = f(x_2)$

IF $|x_2 - x_1| < \text{tolerance}$ THEN

PRINT "Root =", x_2

STOP

END IF

Set $x_0 = x_1$

Set $x_1 = x_2$

UNTIL convergence

END